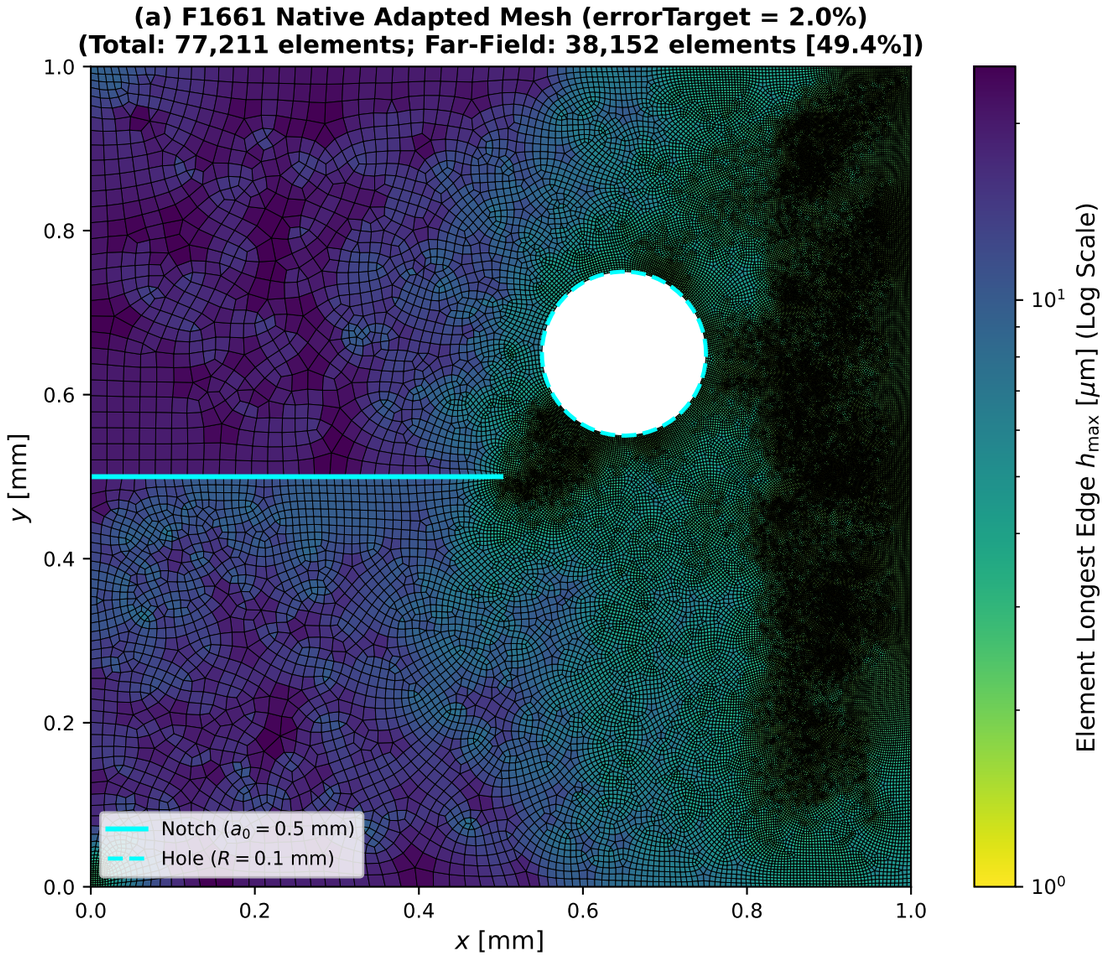
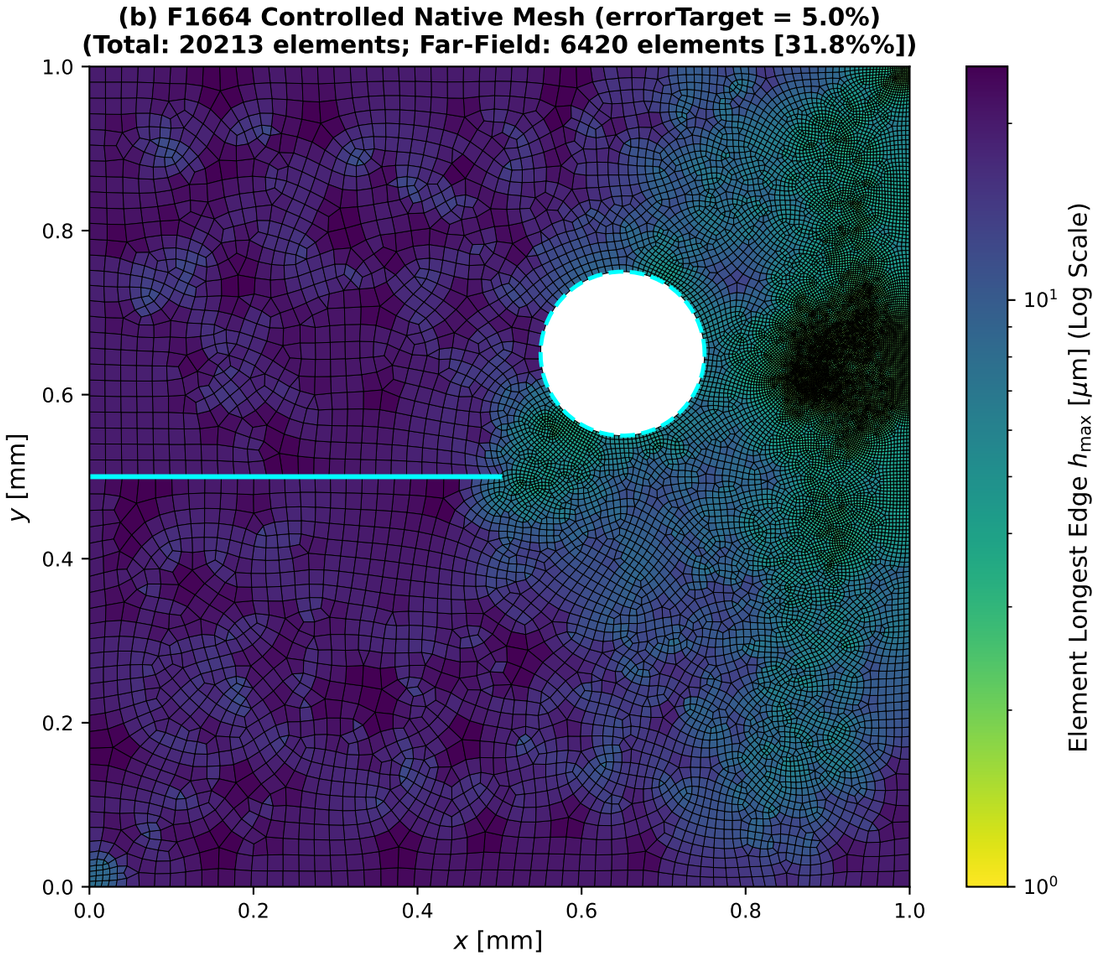
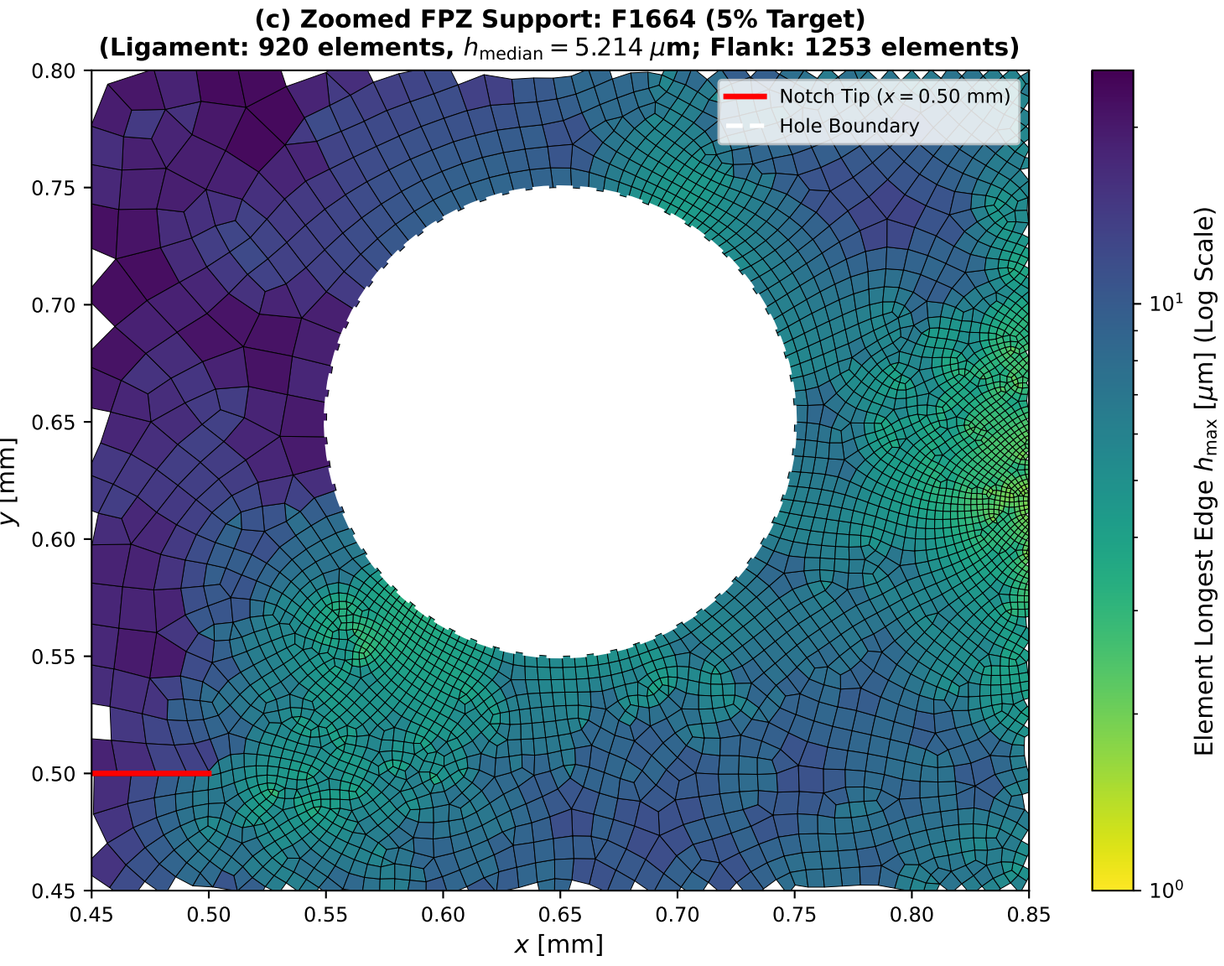
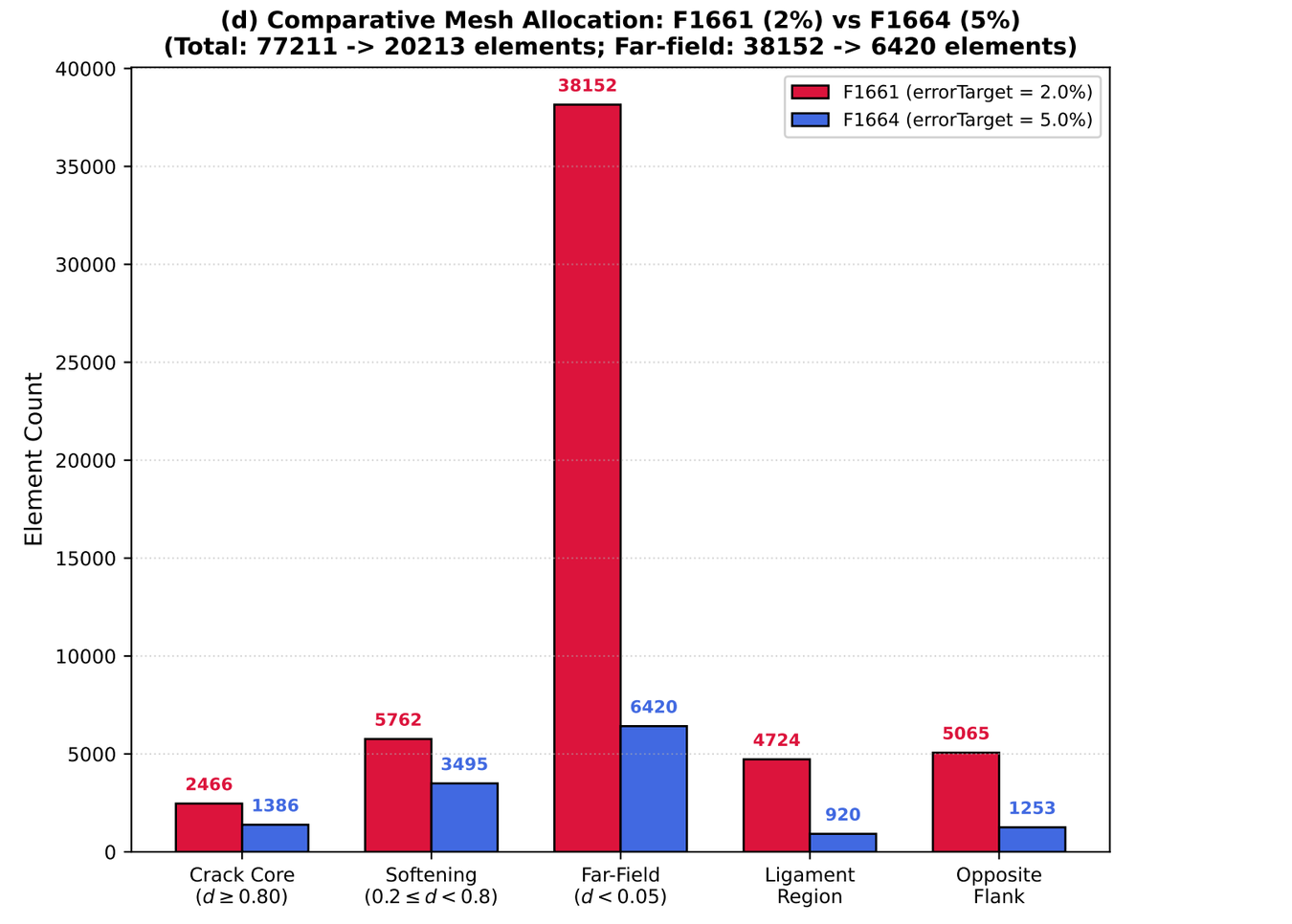

(a) F1661 Native Adapted Mesh (errorTarget = 2.0%)
(Total: 77,211 elements; Far-Field: 38,152 elements [49.4%])
Notch ($a_0 = 0.5$ mm)
Hole ($R = 0.1$ mm)
x [mm]
y [mm]
Element Longest Edge h_{max} [μm] (Log Scale)


(b) F1664 Controlled Native Mesh (errorTarget = 5.0%)
(Total: 20213 elements; Far-Field: 6420 elements [31.8%%])
x [mm]
y [mm]
Element Longest Edge h_{max} [μm] (Log Scale)


(c) Zoomed FPZ Support: F1664 (5% Target)
(Ligament: 920 elements, $h_{median} = 5.214$ μm; Flank: 1253 elements)
Notch Tip ($x = 0.50$ mm)
Hole Boundary
x [mm]
y [mm]
Element Longest Edge h_{max} [μm] (Log Scale)


(d) Comparative Mesh Allocation: F1661 (2%) vs F1664 (5%)
(Total: 77211 -> 20213 elements; Far-field: 38152 -> 6420 elements)
F1661 (errorTarget = 2.0%)
F1664 (errorTarget = 5.0%)
Element Count
2466
1386
5762
3495
38152
6420
4724
920
5065
1253
Crack Core ($d \geq 0.80$)
Softening ($0.2 \leq d < 0.8$)
Far-Field ($d < 0.05$)
Ligament Region
Opposite Flank